Package ‘scanr’

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Type Package

Title Sequential Change-Point Detection with a Rust Backend

Version 0.1.0

Description Detects change points in univariate time series using the SCAN framework with adaptive nonparametric inference. The implementation uses a native Rust backend exposed to R via 'extendr'.

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Archs x64

Contents

`covering_metric` *Segment covering metric*

Description

Computes a weighted segment-covering score in `[0, 1]`.

Usage

```
covering_metric(true_cps, estimated_cps, n)
```

Arguments

`true_cps`	Integer vector of true change points.
`estimated_cps`	Integer vector of estimated change points.
`n`	Number of observations in the series.

Value

Numeric covering score.

`cpd_metrics` *Combined change-point accuracy metrics*

Description

Combined change-point accuracy metrics

Usage

```
cpd_metrics(true_cps, estimated_cps, n, tolerance = 10L)
```

Arguments

`true_cps`	Integer vector of true change points.
`estimated_cps`	Integer vector of estimated change points.
`n`	Number of observations in the series.
`tolerance`	Maximum absolute distance allowed for a tolerant match.

Value

A list with matches, precision, recall, F1, and covering score.

f1_score_cpd — *Tolerant F1 score for change-point detection*

Description

Tolerant F1 score for change-point detection

Usage

```
f1_score_cpd(true_cps, estimated_cps, tolerance = 10L)
```

Arguments

`true_cps`	Integer vector of true change points.
`estimated_cps`	Integer vector of estimated change points.
`tolerance`	Maximum absolute distance allowed for a match.

Value

Numeric F1 score.

ipm_statistic — *Integral probability metric statistic*

Description

Integral probability metric statistic

Usage

```
ipm_statistic(left, right)
```

Arguments

`left`	Numeric vector.
`right`	Numeric vector.

Value

Numeric distance.

match_change_points *Match true and estimated change points*

Description

Greedily matches true and estimated change points by smallest distance, allowing each point to be used at most once.

Usage

```
match_change_points(true_cps, estimated_cps, tolerance = 10L)
```

Arguments

`true_cps`	Integer vector of true change points.
`estimated_cps`	Integer vector of estimated change points.
`tolerance`	Maximum absolute distance allowed for a match.

Value

A data frame with columns `true`, `estimated`, and `distance`.

plot_change_points *Plot detected change points from a scan result*

Description

Plot detected change points from a scan result

Usage

```
plot_change_points(
  x,
  result,
  true_change_points = NULL,
  index = NULL,
  x_label = "Time",
  y_label = "Series",
  title = NULL,
  ...
)
```

Arguments

`x`	Numeric vector.
`result`	A `scanr_result` object.
`true_change_points`	Optional true change points.
`index`	Optional x-axis index with the same length as `x`.
`x_label`	X-axis label.
`y_label`	Y-axis label.
`title`	Plot title.
`...`	Additional arguments passed to `plot()`.

Value

Invisibly returns the plotted data and mapped change points.

`plot_swal_curve` *Plot the SWAL/Wasserstein localization curve for a region*

Description

Plot the SWAL/Wasserstein localization curve for a region

Usage

```
plot_swal_curve(
  x,
  start,
  end,
  x_label = "Time series",
  y_label = "Scaled Wasserstein statistic",
  title = NULL,
  ...
)
```

Arguments

`x`	Numeric vector.
`start`	First observation in the region, using R's one-based indexing.
`end`	Last observation in the region, inclusive.
`x_label`	X-axis label.
`y_label`	Y-axis label.
`title`	Plot title.
`...`	Additional arguments passed to `plot()`.

Value

Invisibly returns the curve data and localized change point.

plot_thresholds — *Plot scan statistics and bootstrap thresholds*

Description

Plot scan statistics and bootstrap thresholds

Usage

```
plot_thresholds(
  result,
  window_size = NULL,
  x_label = "Window start",
  y_label = "Statistic",
  title = NULL,
  ...
)
```

Arguments

`result`	A `scanr_result` object.
`window_size`	Optional window size to plot. Defaults to the first available window.
`x_label`	X-axis label.
`y_label`	Y-axis label.
`title`	Plot title.
`...`	Additional arguments passed to `plot()`.

Value

Invisibly returns the threshold data.

plot_time_series — *Plot a time series with optional change-point markers*

Description

Plot a time series with optional change-point markers

Usage

```
plot_time_series(
  x,
  change_points = NULL,
  true_change_points = NULL,
  index = NULL,
  x_label = "Time",
  y_label = "Value",
  title = "Time series",
  ...
)
```

Arguments

`x`	Numeric vector.
`change_points`	Optional detected change points.
`true_change_points`	Optional true change points.
`index`	Optional x-axis index with the same length as `x`.
`x_label`	X-axis label.
`y_label`	Y-axis label.
`title`	Plot title.
`...`	Additional arguments passed to `plot()`.

Value

Invisibly returns the plotted data and mapped change points.

`plot_vote_scree` *Plot retained change-point count by voting threshold*

Description

Plot retained change-point count by voting threshold

Usage

```
plot_vote_scree(
  result,
  x_label = "Voting threshold",
  y_label = "Number of retained change points",
  title = NULL,
  ...
)
```

Arguments

`result`	A `scanr_result` object.
`x_label`	X-axis label.
`y_label`	Y-axis label.
`title`	Plot title.
`...`	Additional arguments passed to `plot()`.

Value

Invisibly returns the scree data.

`plot_window_votes` *Plot ensemble vote counts for candidate change points*

Description

Plot ensemble vote counts for candidate change points

Usage

```
plot_window_votes(
  result,
  x_label_angle = 45,
  x_label = "Candidate change point",
  y_label = "Window votes",
  title = NULL,
  ...
)
```

Arguments

`result`	A `scanr_result` object.
`x_label_angle`	Rotation angle for x-axis labels.
`x_label`	X-axis label.
`y_label`	Y-axis label.
`title`	Plot title.
`...`	Additional arguments passed to `plot()`.

Value

Invisibly returns the vote data.

`precision_recall_cpd` *Tolerant precision and recall for change-point detection*

Description

Tolerant precision and recall for change-point detection

Usage

```
precision_recall_cpd(true_cps, estimated_cps, tolerance = 10L)
```

Arguments

`true_cps`	Integer vector of true change points.
`estimated_cps`	Integer vector of estimated change points.
`tolerance`	Maximum absolute distance allowed for a match.

Value

Named numeric vector with `precision` and `recall`.

`refine_cusum` *Localize a mean change with a CUSUM statistic*

Description

Localize a mean change with a CUSUM statistic

Usage

```
refine_cusum(x)
```

Arguments

`x`	Numeric vector.

Value

Integer split position.

`refine_wasserstein` *Localize a distributional change with a Wasserstein statistic*

Description

Localize a distributional change with a Wasserstein statistic

Usage

```
refine_wasserstein(x)
```

Arguments

`x`	Numeric vector.

Value

A list with `change_point` and `statistics`.

scan_cpd *Detect change points in a univariate time series*

Description

Detect change points in a univariate time series

Usage

```
scan_cpd(
  x,
  window_sizes = NULL,
  alpha = 0.05,
  n_boot = 400L,
  vote_threshold = 0.5,
  min_window = 15L,
  max_window = NULL,
  block_length = NULL,
  taper = c("tukey", "none"),
  tolerance = NULL,
  random_state = NULL,
  n_jobs = NULL,
  return_all = TRUE,
  change_type = c("distribution", "mean", "var"),
  eps = 1e-12,
  batch_size = 32L
)
```

Arguments

`x`	Numeric vector.
`window_sizes`	Optional positive integer vector of scan window sizes.
`alpha`	Significance level, either as a proportion such as `0.05` or a percentage such as `5`.
`n_boot`	Number of tapered block bootstrap replications.
`vote_threshold`	Minimum normalized ensemble vote score for retained change points.
`min_window`	Minimum default window size when `window_sizes` is `NULL`.
`max_window`	Maximum default window size when `window_sizes` is `NULL`.
`block_length`	Optional tapered block bootstrap block length.
`taper`	Taper shape, either `"tukey"` or `"none"`.
`tolerance`	Distance used to merge nearby candidates across windows.
`random_state`	Optional non-negative integer seed.
`n_jobs`	Optional number of Rust/Rayon worker threads. Defaults to 1. Use `-1` for all detected cores.
`return_all`	Whether to keep per-window diagnostics and raw output.
`change_type`	One of `"distribution"`, `"mean"`, or `"var"`.
`eps`	Small positive value used to avoid division by zero.
`batch_size`	Bootstrap batch size used by the Rust backend.

Value

An object of class scanr_result.

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scan_single_window *Run SCAN for one window size*

Description

Run SCAN for one window size

Usage

```
scan_single_window(
  x,
  window_size,
  alpha = 0.05,
  n_boot = 400L,
  block_length = NULL,
  taper = c("tukey", "none"),
  random_state = NULL,
  change_type = c("distribution", "mean", "var"),
  eps = 1e-12,
  batch_size = 32L
)
```

Arguments

x	Numeric vector.
window_size	Positive integer scan window size.
alpha	Significance level, either as a proportion such as 0.05 or a percentage such as 5.
n_boot	Number of tapered block bootstrap replications.
block_length	Optional tapered block bootstrap block length.
taper	Taper shape, either "tukey" or "none".
random_state	Optional non-negative integer seed.
change_type	One of "distribution", "mean", or "var".
eps	Small positive value used to avoid division by zero.
batch_size	Bootstrap batch size used by the Rust backend.

Value

An object of class scanr_window_result.

`swal_statistic` *Local SCAN/Wasserstein split statistic*

Description

Local SCAN/Wasserstein split statistic

Usage

```
swal_statistic(x, change_type = c("distribution", "mean", "var"))
```

Arguments

`x`	Numeric vector.
`change_type`	One of `"distribution"`, `"mean"`, or `"var"`.

Value

Integer split position.

`wasserstein_statistic` *One-dimensional Wasserstein distance*

Description

One-dimensional Wasserstein distance

Usage

```
wasserstein_statistic(left, right)
```

Arguments

`left`	Numeric vector.
`right`	Numeric vector.

Value

Numeric distance.

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